INCH-POUND

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DETAIL SPECIFICATION

OXYGEN CANDLE, SODIUM CHLORATE

This specification is approved for interim use by Naval Sea Systems Command. Other activities in the Department of Defense may use this interim amendment or may continue using MIL-DTL-32789.

1. SCOPE

1.1 Scope. This specification covers sodium chlorate oxygen candles and ignition devices for use only in Navy standard candle furnaces to supply emergency breathing oxygen in submarines.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

PPP-C-96 - Cans, Metal, 28 Gage and Lighter

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-117 - Bags, Heat-Sealable

MIL-DTL-11195 - Enamel, Lusterless, Fast Dry, VOC Compliant, (for Use on Ammunition and Other Metals)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Military Marking for Shipment and Storage

MIL-STD-3010 - Test Procedures for Packaging Materials and Containers

(Copies of these publications are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this publication are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). This publication is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this publication are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D5330/D5330M - Standard Specification for Pressure-Sensitive Tape for Packaging, Filament-Reinforced

(Copies of this publication are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS-STD-595 - Colors Used in Government Procurement

(Copies of this publication are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Retrospective analysis. For each production month, production samples shall be reserved for analysis at a Government facility in accordance with 4.3 (see 6.2).

3.3 Conformance inspection. For each production lot (see 4.4.2), a sample shall be inspected at the manufacturer's facility in accordance with 4.4 (see 6.2).

3.4 Material. The sodium chlorate oxygen candle shall consist primarily of a mixture of sodium chlorate ($NaClO_3$) and iron powder or steel wool pressed or cast into a cylindrical shape sufficient in size to generate the required volume of oxygen (see 3.7.3). Barium peroxide (BaO_2) may be used up to a mass percentage of 5 percent. Non-toxic, oxygen-compatible binders may be used (see 6.3.6).

3.4.1 Prohibited materials. The sodium chlorate oxygen candle shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs.

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3.5 Candle design and construction.

3.5.1 Dimensions. The diameter shall be not less than 6⅛ inches (15.6 centimeters). The maximum diameter shall be such that the candle fits in the required can (see 3.9.1). The length shall be not less than 11⅜ inches (28.9 centimeters) nor greater than 11⅝ inches (29.5 centimeters) (see 4.5.1).

3.5.2 Consistency. The appearances of all candles shall be consistent to aid visual detection of irregularities, such as contaminants that may adversely affect candle performance (see 4.5.2.1). Minor variations in the appearance of the ignition zone of the candles shall be permitted.

3.5.3 Chips and loose material. The canned candle shall not produce chips and loose material that are more than 1 percent of the candle's total weight (see 4.5.2.2).

3.5.4 Cracks. Longitudinal cracks in the candle body shall not extend the full length of the candle, and radial cracks shall not exceed the length of the radius or cut the circumference in two or more places (see 4.5.2.3).

3.5.5 Ignition zone. A disk or zone of candle material of special formulation, centered at the top of the candle and flush with the surface, shall be an integral part of the candle. The ignition zone shall not be misaligned by more than ⅜ inch (1 centimeter) from the center of the top and shall be tightly retained. The ignition zone may contain minor chips and cracks (see 4.5.1).

3.6 Ignition device. One ignition device (igniter) shall be provided with each candle. The igniter shall be a steel rod, 0.135 to 0.148 inch (3.43 to 3.76 millimeters) in diameter and 6 inches (15.24 centimeters) in length, with one end pointed for insertion through a fitting and the other end with a ½-inch (1.27-centimeter) diameter head for application of amorphous red phosphorous (see 6.3.8). The shaft shall be free of burrs or other manufacturing blemishes that would impede the igniter from passing freely through the fitting. The igniter shall be coated on the head with no less than 0.62 grain (0.04 gram) and no more than 1.5 grain (0.1 gram) of amorphous red phosphorus (see 4.5.3.8).

3.7 Performance characteristics.

3.7.1 Burn time. The burn time shall be not less than 40 minutes nor more than 60 minutes for complete burn of one candle when burned in a Navy standard candle furnace or alternate approved furnace (see 6.2) at an ambient temperature of 75±11 °F (24±6 °C) (see 4.6.1.4.1).

3.7.2 Burn rate. During the burn time, the maximum instantaneous burn rate, measured as oxygen flow, shall be not greater than five times the average burn rate. The candle shall have a clearly observable end of burn (see 4.6.1.4.2).

3.7.3 Oxygen content. Each candle shall provide no less than 118 cubic feet of oxygen (3341 liters) at standard temperature and pressure (STP). STP shall be 32 °F (0 °C) and 14.69 pounds per square inch absolute (760 millimeters of mercury) (see 4.6.1.4.3).

3.7.4 Ignition requirement. The candle shall successfully ignite in the Navy furnace with an ignition device (see 3.6) that is passed through the fitting in the center of a lid of a candle furnace, pressed against the center of the top of a candle (see 3.5.5), and rotated by hand from outside of the furnace. No other energy source shall be used to initiate the reaction (see 4.6.1.4.4).

3.8 Byproducts of burning.

3.8.1 Water vapor. Water, when calculated based on the total weight of water contained in the volume of oxygen sampled, shall not exceed 4.4 grains of water per cubic foot of oxygen (10 milligrams of water per liter of oxygen) when averaged over the entire burn. This corresponds to a dew point of approximately 50 °F (10 °C) (see 4.6.1.5.1).

3.8.2 Total chlorine. Chlorine and other oxidizing chlorine compounds (such as chlorine dioxide), when calculated based on the total volume of chlorine contained in the total volume of oxygen under the same temperature and pressure conditions, shall not exceed 2.5 parts per million (ppm) when averaged over the entire burn (see 4.6.1.5.2).

3.8.3 Carbon monoxide. Carbon monoxide, when calculated based on the total volume of carbon monoxide contained in the total volume of oxygen under the same temperature and pressure conditions, shall not exceed 25 ppm when averaged over the entire burn (see 4.6.1.5.3).

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3.9 Containers. When tested in accordance with 4.5.3.5, the candle and ignition device containers shall conform to the requirements of 3.9.1 through 3.10.2.

3.9.1 Candle can. Each candle shall be unit packaged in a cylindrical metal can in accordance with PPP-C-96, type I, class 1, and the following requirements:

a. The inside diameter shall be 6 9/16±1/16 inches (16.7±0.2 centimeters).

b. The can shall be 11 7/8±1/8 inches (30.2±0.3 centimeters) in length.

c. The material for the can and body ends shall be a No. 135 electrolytic tin plate with a thickness of 28 gauge (0.0149 inch or 0.38 millimeter).

d. Side seams shall be welded in a continuous and uniform manner through the entire seam length. The top and bottom shall be double-seamed. The tear strip location may be soldered.

e. Both ends shall be reinforced by raised circular rings.

f. The key shall be made in accordance with PPP-C-96 and shall be affixed to the can end cover so that it is easily removed.

g. The candle shall be inserted within the can so that the ignition end of the candle is packaged at the end opposite the tear strip.

h. The canned candles shall be successfully opened using the tear strip and key when tested in accordance with 4.5.3.6.

3.9.1.1 Coatings. The interior of the can shall be in accordance with PPP-C-96. The exterior of the can shall be provided with a uniform and continuous finish coat of enamel in accordance with MIL-DTL-11195, with the color approximating color number 34108 of SAE AMS-STD-595. Prior to the finish coat, all exterior surfaces of the can shall be cleaned and provided a suitable surface to ensure a tightly adhering finish coat. The finish coat shall be free from cracks, scratches, or other defects or irregularities that may affect the adhesive and corrosion-resistant properties of the coating. All cracks, scratches, or other paint defects or irregularities in the finish coat after the canning process of the oxygen candle shall be retouched by hand with the same paint.

3.9.1.2 Protective pads. Non-organic, oxygen-compatible packaging material in the form of 6.25-inch (15.9-centimeter) nominal diameter pads shall be used on each end of the candle in the can to protect the candle from abrasion. A hole shall be cut in the center of the pad over the ignition end to prevent contact of the ignition zone with the pad. Use of the same igniter-end pad for the opposite end of the candle is acceptable, or alternatively, a solid pad may be used.

3.9.1.3 Canned candle integrity. The candle can after sealing shall prevent intrusion of any oils, moisture, or other contaminants (see 4.6.2).

3.9.2 Ignition device container.

3.9.2.1 Ignition device envelope. Each ignition device shall be packed in a heat-sealed metal foil envelope in accordance with MIL-DTL-117, type I, class E, style 1, with outside dimensions of 2⅛ by 7⅜ inches (5.4 by 18.7 centimeters).

3.9.2.2 Ignition device tube. The enveloped ignition device shall be enclosed in an aluminum alloy tube with nominally sized dimensions of 1 inch (2.5 centimeters) in diameter and 7 inches (17.8 centimeters) in length and a screw cap. Alternatively, a slip-neck-type aluminum tube is acceptable.

3.9.2.3 Ignition device tube cap sealing. The screw cap or slip neck shall be sealed with type III or IV pressure-sensitive tape in accordance with ASTM D5330/D5330M with water-insoluble adhesive.

3.9.3 Assembly. Each candle assembly shall consist of a candle (unit) comprised of material as specified in 3.4 and stored in a can as specified in 3.9.1. Each can shall contain the candle with protective pads on each end (see 3.9.1.2) and an ignition device (see 3.6) stored in an attached ignition device container (see 3.9.2). The ignition device container shall be attached to the candle can by two bands of type III or IV pressure-sensitive tape in accordance with ASTM D5330/D5330M with water-insoluble adhesive that completely circles the circumference of the can and igniter container (see 4.5.3.3).

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3.10 Marking. Each can shall be marked in accordance with MIL-STD-129 and as specified in 3.10.1 through 3.10.2 (see 4.5.3.7).

3.10.1 Candle can. Each candle can label shall include the following caution marking:

CAUTION
STORE IN CLEAN, COOL, DRY PLACE
DO NOT OPEN CAN UNTIL READY FOR USE
CANDLE MAY IGNITE IF TEMPERATURE EXCEEDS 400 °F (204 °C)
NEVER ALLOW WATER, OIL, GREASE, OR COMBUSTIBLES TO CONTACT CANDLE

3.10.2 Ignition device container. Each ignition device container label shall include the CAGE code, noun name, part number, quantity, and “less than 0.1 gram red phosphorous (flammable)”.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Retrospective analysis (see 4.3).

c. Conformance inspection (see 4.4).

4.2 First article inspection. When required (see 3.1), a sample shall be subjected to first article inspection. First article inspection shall consist of the examinations of 4.5 and tests of 4.6 as specified in table I at a location acceptable to the Government (see 6.2).

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TABLE I. First article inspection, retrospective analysis, and conformance inspection.

Characteristic	First article inspection	Retrospective analysis	Conformance inspection	Requirement	Verification
Material	X	-	-	3.4	4.5.3.1
Prohibited materials	X	-	-	3.4.1	4.5.3.2
Candle design and construction:					
Dimensions	X	-	X	3.5.1	4.5.1
Consistency	X	-	X	3.5.2	4.5.2.1
Chips and loose material	X	-	X	3.5.3	4.5.2.2
Cracks	X	-	X	3.5.4	4.5.2.3
Ignition zone	X	-	X	3.5.5	4.5.1
Ignition device	X	-	-	3.6	4.5.3.8
Performance characteristics:					
Burn time	X	X	X	3.7.1	4.6.1.4.1
Burn rate	X	X	X	3.7.2	4.6.1.4.2
Oxygen content	X	X	X	3.7.3	4.6.1.4.3
Ignition requirement	X	X	X	3.7.4	4.6.1.4.4
Byproducts of burning:					
Water vapor	X	X	X	3.8.1	4.6.1.5.1
Total chlorine	X	X	X	3.8.2	4.6.1.5.2
Carbon monoxide	X	X	X	3.8.3	4.6.1.5.3
Containers:					
Candle can	X	-	-	3.9.1	4.5.3.5
Successful opening	X	-	-	3.9.1	4.5.3.6
Coatings	X	-	-	3.9.1.1	4.5.3.5
Protective pads	X	-	-	3.9.1.2	4.5.3.5
Canned candle integrity	X	-	X	3.9.1.3	4.6.2
Ignition device container	X	-	-	3.9.2	4.5.3.5
Assembly	X	-	-	3.9.3	4.5.3.3
Tape	X	-	-	3.9.3	4.5.3.4
Marking	X	-	-	3.10	4.5.3.7

4.2.1 Sampling. Sampling for first article inspection shall be as follows:

a. A total of 12 candles in the specified cans (see 3.9.1) shall be submitted for first article inspection.

b. A total of 24 igniters in the specified containers (see 3.9.2) shall be submitted for first article inspection. 12 igniters shall be assembled with the 12 candle cans (see 3.9.1), and 12 extra igniters containerized but not attached to candle cans shall be supplied.

c. For all candle and ignition device containers submitted for first article inspection, additional labels shall be applied to all containers stating, “FIRST ARTICLE INSPECTION”.

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d. No fewer than 10 candles of the 12 supplied for first article inspection shall be tested. Should one candle fail any of the critical tests (see 4.2.2.1), the remaining two candles shall be tested.

4.2.2 Classification of defects for first article inspection.

4.2.2.1 Critical defects. Failures of any requirement during inspection tests are classified as critical defects unless listed as minor (see 4.2.2.2).

4.2.2.2 Minor defects. Failure of the following requirements are classified as minor defects:

a. Assembly (see 3.9.3 and 4.5.3.3).

b. Water vapor (see 3.8.1 and 4.6.1.5.1).

c. Marking (see 3.10 and 4.5.3.7).

4.3 Retrospective analysis. Retrospective analysis shall consist of the tests of 4.6 as specified in table I at a location acceptable to the Government (see 6.2 and 6.3.2).

4.3.1 Sampling. Sampling for retrospective analysis of production shall be as follows:

a. A total of eight candles in the specified cans (see 3.9.1) shall be submitted for retrospective analysis from the lots produced in each month during which production occurs. The eight candles shall represent as many different lots produced during that month as possible.

b. A total of 16 igniters in the specified containers (see 3.9.2) shall be submitted for retrospective analysis. Eight igniters shall be assembled with the eight candle cans (see 3.9.1), and eight extra igniters containerized but not attached to candle cans shall be supplied.

c. For all candle and ignition device containers submitted for retrospective analysis, additional labels shall be applied to all containers stating, “RETROSPECTIVE ANALYSIS”.

4.4 Conformance inspection. Conformance inspection shall consist of the examinations of 4.5 and tests of 4.6 as specified in table I (see 6.2 and 6.3.3).

4.4.1 Sampling. For examinations specified in 4.5 and the leak test specified in 4.6.2, a random sample of chlorate candles shall be selected from each production lot (see 4.4.2) offered in accordance with ASQ/ANSI Z1.4 at inspection level III and the Acceptance Quality Limit (AQL) specified (see 6.2). For the burn test specified in 4.6.1, a random sample of chlorate candles shall be selected from each production lot offered in accordance with ASQ/ANSI Z1.4 at inspection level S-3 and the AQL specified (see 6.2). In the event that multiple ovens are used, samples shall be taken as equally as possible from each oven totaling the amount required for lot inspection. Where a lot involves more than one chemical mix or batch, the samples shall be chosen at random and be representative of as many mixes or batches as possible. The number of candles required by ASQ/ANSI Z1.4 is the minimum number of candles to be tested. More candles may be inspected than the minimum.

4.4.2 Production lot. A lot shall be defined as one oven load. The total shall not exceed the maximum drying capacity of a single drying oven if multiple ovens are used.

4.5 Examinations. Dimensional and visual examinations shall be conducted on the required number of candles (see 4.4.1) after removal from the drying oven(s) and cooling until at a safe-to-handle temperature.

4.5.1 Dimensional examination. The dimensions of the samples shall be examined as follows:

a. The dimensions of the samples shall be verified for conformance to 3.5.1.

b. Dimensional examination shall include measurement of the central location of the ignition zone (see 3.5.5). A custom gauge capable of producing a pass/fail result rather than a numerical value may be used to verify conformance.

c. Measurements of candle length, diameter, ignition zone centering, and ignition device shall be acquired using calibrated instruments with precisions of 1/16 inch (0.1 centimeter) or better.

4.5.2 Visual examination. Samples shall be visually inspected for consistency, chips, and cracks as specified in 4.5.2.1 through 4.5.2.3.

4.5.2.1 Consistency. Sampled candles shall be visually examined for consistent appearance (see 3.5.2).

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4.5.2.2 Chips and loose material. For first article inspection, candles shall be examined for chips and loose material (see 3.5.3) after canning and leak testing (see 4.6.2). For conformance inspection, candles shall be examined for chips and loose material after removal from oven and cooling to a safe-to-handle temperature.

a. All loose material or chips (excluding the top and bottom padding if the candle was canned prior to examination) shall be transferred to a suitable pre-tared container and weighed.

b. The weight of loose chips shall be measured using a calibrated scale with a capacity of 30 pounds (14 kilograms) and a resolution of 0.05 pound (20 grams) or better. The same scale may be used for loose material and total candle weight.

4.5.2.3 Cracks. Sampled candles shall be examined for the presence of cracks (see 3.5.4).

4.5.3 Additional examinations for first article inspection.

4.5.3.1 Material verification. The candles shall be verified during first article inspection for conformance to the requirements of 3.4 as specified (see 6.2).

4.5.3.2 Prohibited materials. Prohibited materials shall be verified as specified (see 3.4.1 and 6.2).

4.5.3.3 Verification of assembly. The assembly of the candle and igniter shall conform to 3.9.3 and be verified visually.

4.5.3.4 Tape. Tape requirements shall be verified as specified (see 6.2).

4.5.3.5 Containers. The candle and igniter containers shall conform to 3.9 and be verified visually. The presence of the protective pads, tape, adhesive labels, and coating shall be verified visually.

4.5.3.6 Tear strip and key. The candle shall be verified to open successfully using the tear strip and key (see 3.9.1).

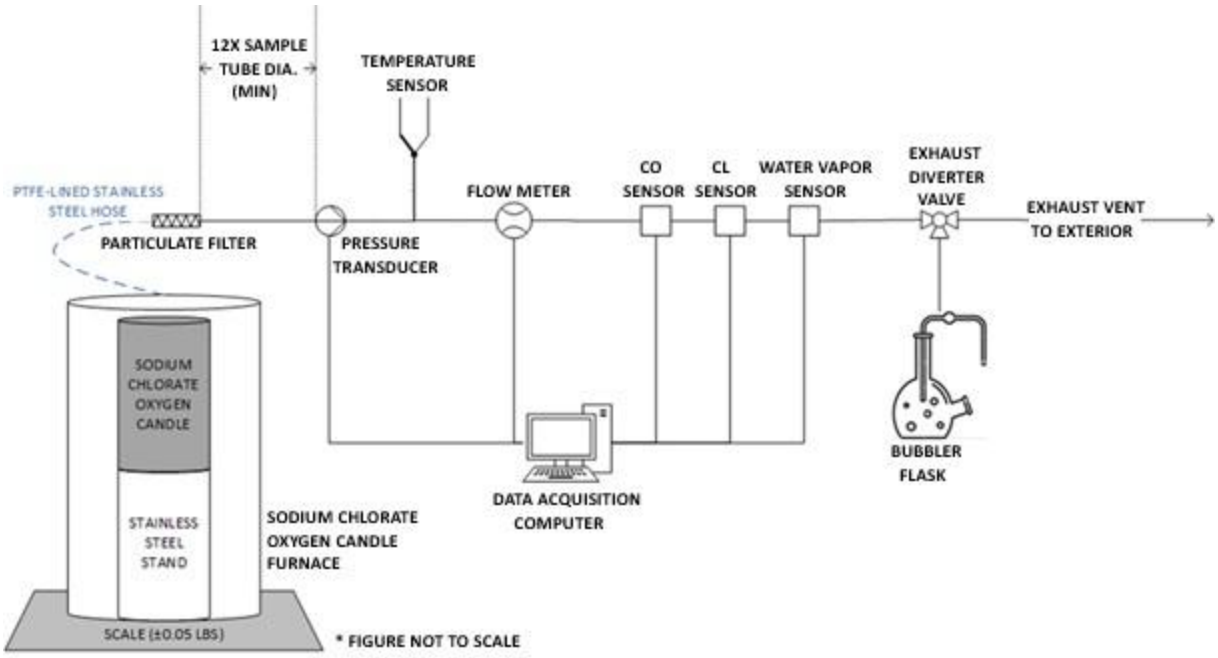
4.5.3.7 Marking. The marking of the candle and igniter containers shall conform to 3.10 and be verified visually.

4.5.3.8 Ignition device. The ignition device shall be examined to verify conformance to dimensional requirements, to confirm that it is free of manufacturing defects, and to ensure the presence of phosphorous on head as specified in 3.6. The red phosphorous coating shall be verified visually to ensure that the entire head of the igniter is covered. The length shall be verified using a calibrated device with precision of at least 1/16 inch or 0.1 centimeter. The igniter shaft's diameter shall be verified using a calibrated caliper or micrometer with precision of at least 0.01 inch or 0.25 millimeter. Candles undergoing examination shall only utilize ignition devices shipped by the same manufacturer.

4.6 Tests.

4.6.1 Burn test. Test candles shall be burned singly in a Navy standard candle furnace and ignited using the standard ignition device provided. The burn test rig shall adhere to the requirements on figure 1 and in table II. To burn single candles, an unpainted stainless-steel cylindrical stand 6⅜ inches (16.2 centimeters) in diameter and 11½ inches (29.2 centimeters) in height shall be first inserted into the furnace's inner basket to support the test candle. Any modifications to or substitutions for the candle furnace require approval (see 6.2) prior to testing.

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The test rig layout shall strictly adhere to all manufacturer's instructions and recommendations for proper use, operation, and location of the instrumentation equipment. Test hardware, including the candle furnace and filters, shall be cleaned frequently to remove residue from previous candle burns. Buildup of residue could negatively affect test data. Note: This figure does not preclude the installation of a pre-filter upstream of the oxygen flow meter to mitigate the tendency for particulate buildup in the test rig. Deviations from the test rig layout or specific components require approval (see 6.2).

FIGURE 1. Schematic diagram of candle test apparatus.

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TABLE II. Candle test apparatus components.

Component	Description
Standard submarine candle furnace	Stainless-steel candle furnace diameter: 16 inches (40.6 centimeters), height: 29 inches (73.7 centimeters), weight: 86 pounds (33.9 kilograms), with dual fiberglass filters. Stainless-steel cylindrical insert, 6⅜ inches (16.2 centimeters) in diameter and 11½ inches (29.2 centimeters) in height, to support the test candle, 0 to 15 pounds per square inch gauge (psig) (0 to 1 bar) pressure gauge, maximum operating pressure 10 psig (0.6 bar), relief valve setpoint 13±2 psig (0.8±0.01 bar).
Laboratory scale	Calibrated laboratory quality scale. Minimum range of 0 to 30 pounds (0 to 15 kilograms) with resolution of 0.05 pound (20 grams) or better.
Pressure transducer	Pressure transducer of sufficient range (see 6.3.7.1) shall be calibrated and within calibration period. Accuracy for pressure reading given by the manufacturer shall be ±0.5 percent or better.
Flow meter	Mass flow meter or turbine meter of sufficient range (see 6.3.7.2). Flow meter shall be calibrated and within calibration period.
Thermocouple or thermistor	Standard commercial. Thermocouple or thermistor shall be calibrated and within calibration period.
Data acquisition computer	Device for recording and manipulating sensor data. Compatible with all instrumentation.
In-line particulate filter	Standard commercial sufficient for collecting salt particulates emitted from candle that would affect flow and sensor responses. Filter shall be cleaned or replaced each day of candle testing.
Water vapor sensor	Calibrated hygrometer of sufficient range (see 6.3.7.3) capable of reporting dew point or milligrams of water per liter.
Chlorine measurement	Electrochemical or infrared device of sufficient range (see 6.3.7.4). Manufacturer's sensor cross-sensitivity data shall indicate that the sensor shall have equal or greater sensitivity to chlorine dioxide (ClO_2). Calibration with chlorine dioxide is not required. Resolution of 0.1 ppm or better. The sensor shall be calibrated and zeroed each day of candle testing.
Carbon monoxide measurement	Electrochemical or infrared device of sufficient range (see 6.3.7.5). Resolution of 1 ppm or better. The sensor shall be calibrated and zeroed each day of candle testing.
Bubbler flask and associated tubing	Commercial glass laboratory flask. Tubing may be any compatible material that is positively retained inside the flask.
Exhaust diverter valve	Two-position ball (three-way) valve.
Calibration gas for chlorine sensor	Certified chlorine gas mixture at concentration recommended by sensor manufacturer. Bottle shall be within certification period provided by vendor.
Calibration gas for carbon monoxide	Certified carbon monoxide gas mixture at concentration recommended by sensor manufacturer. Bottle shall be within certification period provided by vendor.
Zero air calibration gas	Certified zero gas. Bottle shall be within certification period provided by vendor.

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4.6.1.1 Tests preceding burn test. Candles for first article inspection (see 4.2) and retrospective analysis (see 4.3) shall be canned (see 3.9.1), leak tested (see 4.6.2), assembled (see 3.9.3), and marked (see 3.10) prior to burn testing.

4.6.1.2 Exemption to canning for conformance inspection. Candles for conformance inspection (see 4.4) may be canned prior to burn testing. Candles sampled for burn testing shall be maintained at a room temperature of 75±11 °F (24±6 °C) for a period of at least 4 hours prior to testing.

4.6.1.3 Burn test instrumentation. The burn test instrumentation shall monitor the candle output over the full duration of the burn. Straight, hard tubing with a length of at least 12 times the diameter shall be installed upstream of instrumentation, as shown in figure 1. The byproduct levels and oxygen flow rate shall be monitored throughout the burn. The average concentration of byproducts in ppm for the full duration of the burn shall be calculated as specified in 4.6.1.5. Sampling rates for all parameters shall be at least every 5 seconds and be sampled consistently over the entire burn time. If the instantaneous value of any byproduct of burning produced during this test meets or exceeds the maximum range of the selected sensor, another candle from the lot shall be tested as a replacement without the lot being considered a failure.

4.6.1.4 Performance characteristics.

4.6.1.4.1 Burn time. The length of time from ignition to the end of oxygen flow shall conform to 3.7.1. The start of burn time shall be determined by the first value of positive flow by the flow meter. The end of burn time is indicated by no flow through the bubbler for a period of 5 seconds as judged by the test operator. Flow to the bubbler shall be started by shifting the exhaust diverter from the “EXHAUST” to the “BUBBLER” position when the flow meter indicates a flow of 20 standard liters per minute or less. The bubbler shall be constructed of a standard laboratory flask filled with enough water to cover the bottom of the inlet tube. The inlet tube for oxygen shall be positively retained inside the flask with the bottom submerged 1 to 2 inches beneath the water’s surface in the flask. Immediately after confirmation of the end of oxygen flow, the valve shall be returned to the “EXHAUST” position to stop any water from being drawn up the bubbler tube as the system cools, and the burn timer shall be stopped. Burn time shall be measured to the nearest second.

4.6.1.4.2 Burn rate. The maximum instantaneous burn rate shall conform to 3.7.2.

4.6.1.4.2.1 Oxygen flow. The oxygen gas flow shall be monitored throughout the entire burn with a properly calibrated flow device suitable for oxygen flow. Flow measurements shall be generated at least every 5 seconds (see 3.7.3).

4.6.1.4.2.2 Flow rate units. Flow rate data points shall be converted to standard liters per minute (see 3.7.3). If a flow device is used that automatically converts flow data to standard temperature and pressure, this step is unnecessary.

4.6.1.4.2.3 Average burn rate. The average burn rate shall be determined using the flow rate data obtained during the entire candle burn. The average flow rate shall be determined by summing the individual standardized flow rate data values and dividing by the total number of data values.

4.6.1.4.2.4 Maximum flow rate. The maximum rate shall be determined from the standardized flow rate data.

4.6.1.4.2.5 Maximum flow ratio. The maximum flow to average flow ratio shall be calculated as the maximum flow divided by the average rate. The flow ratio shall conform to 3.7.2.

4.6.1.4.3 Oxygen content. The total calculated generated oxygen shall conform to 3.7.3 and shall be conducted in accordance with the following requirements:

a. The original weight of the candle shall be determined with a precision of 0.05 pound (20 grams) or better.

b. The total weight of the burnt candle shall be determined by the loss in weight of the candle during the burn test using a scale with a precision of 0.05 pound (20 grams) or better.

c. The volume of oxygen at STP shall be determined by the following formula:

$$Volume\ of\ oxygen\ (in\ standard\ liters) = 317.8 \times change\ in\ weight\ (in\ pounds)$$

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d. A second value for the total oxygen production shall be computed by multiplying the average burn rate (see 4.6.1.4.2.3) and the burn time (see 4.6.1.4.1).

4.6.1.4.4 Ignition requirement. Proper ignition shall conform to 3.7.4. Ignition shall be accomplished by gently lowering the igniter through the igniter fitting until it contacts the candle surface. While pressing the head of the igniter onto the ignition zone, the head shall be rapidly rotated approximately one half-turn. A faint "hiss" sound or vibration in the igniter may be perceived, and the igniter may sink approximately ¼ inch (approximately ½ centimeter). The ignition device shall be held in place for approximately 5 seconds to allow the chemical reaction to start and then lifted until the head of the igniter shaft impacts the underside of the igniter holder. If the igniter is not lifted after ignition, there is risk of the reaction consuming the igniter and energetic emissions from the fitting. An individual ignition device may be turned by hand twice if the first turn did not cause ignition. If a second igniter is necessary, after ensuring the reaction has not started, the furnace lid and first igniter shall be carefully removed, and a second, fresh igniter shall be inserted.

4.6.1.5 Byproducts of burning.

4.6.1.5.1 Water vapor. The average water vapor shall conform to 3.8.1. Water vapor shall be determined by analytical techniques properly calibrated per supplier requirements. The reported water vapor average content shall be a weighted average based on the sum of the products of the water vapor content multiplied by the oxygen flow rate at the same point of time divided by the total oxygen flow. Alternatively, the average dew point may be computed in place of the average mass of water as a weighted average. The results may be presented as mass of water per volume of gas or as dew points.

4.6.1.5.2 Total chlorine. The average chlorine and other oxidizing chlorine compounds shall conform to 3.8.2. The values of chlorine and other oxidizing chlorine compounds (such as chlorine dioxide) shall be determined by analytical techniques properly calibrated using a suitable chlorine gas standard per supplier requirements. The reported chlorine average concentration shall be a weighted average based on the sum of the products of the chlorine concentration multiplied by the oxygen flow rate at the same point of time divided by the total oxygen flow.

4.6.1.5.3 Carbon monoxide. The average carbon monoxide shall conform to 3.8.3. The values of average carbon monoxide shall be determined by analytical techniques properly calibrated using a suitable carbon monoxide gas standard per supplier requirements. The reported carbon monoxide average concentration shall be a weighted average based on the sum of the products of the carbon monoxide concentration multiplied by the oxygen flow rate at the same point of time divided by the total oxygen flow.

4.6.2 Canned candle leak test. Canned candle leak tests shall be conducted on the required number of candles (see 4.4.1) after canning (see 3.9.1). A sample of canned candles shall be inspected for leakage (see 3.9.1.3). Samples submitted shall be leak tested by method 5009 of MIL-STD-3010 using the hot water technique. Evidence of a steady stream or a recurring succession of bubbles from any surface shall be cause for rejection. If more than the allowed number of leak test failures are found, all the remaining canned candles of the lot shall be leak tested. All canned candles that fail the leak test shall be disposed of.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The chlorate candle is intended to generate breathing oxygen for U.S. Navy submarine personnel. Current stock of sodium chlorate oxygen candles may be used until depleted; future requisitions for sodium chlorate oxygen candles should refer to the current version of this specification. Sodium chlorate oxygen candles are form, fit, and function replacements for prior revisions.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Relevant information for first article inspection (see 3.1, 4.2, and 6.3.1).

c. Relevant information for retrospective analysis (see 3.2, 4.3, and 6.3.2).

d. Relevant information for conformance inspection (see 3.3, 4.4, and 6.3.3).

e. Requirement for prohibited materials verification (see 3.4.1 and 4.5.3.2).

f. Requirements for obtaining approval of modifications to or substitutions for the candle furnace (see 3.7.1 and 4.6.1).

g. For conformance inspections (see 4.4.1), an AQL of 1.0 for examinations (see 4.5) and leak tests (see 4.6.2), and an AQL of 2.5 for burn testing (see 4.6.1).

h. Requirements for obtaining approval of deviations from the test rig layout or specific components (see figure 1).

i. Requirements for material certification as provided by material vendors (see 4.5.3.1).

j. Requirements for documentation of tape conformance (see 4.5.3.4).

k. Packaging requirements (see 5.1).

l. Requirements for disposition of lots failing conformance testing or requesting deviations from test results not meeting requirements (see 6.3.3.1).

6.3 Technical guidance.

6.3.1 First article inspection. First article inspection (see 3.1) should be required when:

a. A vendor has not been in continuous production of sodium chlorate oxygen candles for the Navy for a period of 6 months or more.

b. A change in manufacturing location has occurred.

c. A change in manufacturing or test equipment has been requested by the vendor.

d. DELETED.

6.3.1.1 Failure of first article inspection. If any candles submitted for first article inspection fail one or more critical tests (see 4.2.2.1), the vendor may be required to submit a full set of candles for another first article inspection. If any candles submitted for first article inspection fail one or more minor tests (see 4.2.2.2), the vendor should submit a report with a correction plan to address failures to the Technical Authority (see 6.5).

6.3.1.2 Location of first article inspection. The Government may require that first article inspection be performed at a Government facility or at the vendor's facility. If first article inspection is performed at a Government laboratory, the candle cans should be leak tested by the vendor before shipping. Regardless of test location, a representative from the Technical Authority should be invited to witness. If performed at a Government laboratory, the vendor may request to be present during the first article inspection.

6.3.2 Retrospective analysis. The Government reserves the right to require, waive, or modify periodic testing during production and the right to the selection of test samples and test sites as specified in 3.2.

6.3.2.1 Failure of the retrospective analysis. Failure to meet testing or examination requirements may be cause for rejection of lot(s) and tightened conformance inspection (see 3.3 and 4.4) requirements. Tightening of conformance inspection may be done by raising the sample size requirement per ASQ/ANSI Z1.4.

6.3.2.2 Location of retrospective analysis. The acquisition activity should direct the vendor to provide samples for retrospective analysis to a Government laboratory specified by the Technical Authority. The candle cans should be leak tested at the vendor's facility before shipping to the Government laboratory.

6.3.3 Conformance inspection. Conformance testing is a continuous process to be performed for each lot (see 4.4.2) during production.

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6.3.3.1 Failure of conformance inspection. If the number of candle failures exceeds those required by 4.4.1, the lot may be rejected. If a lot of production candles fails conformance inspection, the vendor may request direction from the Government to dispose of all or part of the lot or a waiver to rework the lot rather than dispose of it, contingent upon approval from the Technical Authority. All requirements of conformance inspection apply to the reworked lot. No additional candles should be selected from the lot to substitute for a failed candle from the original sample.

6.3.3.2 Location of conformance inspection. Conformance testing is to be performed at a vendor-owned facility, typically the factory.

6.3.4 Oxygen safety. Candles generate high concentrations of oxygen, approaching 100 percent. The vendor must be aware of hazards associated with solid oxidizers and high-concentration oxygen atmospheres. Useful information on oxygen system safety may be found in ASTM MNL36, ASTM G63 (non-metals), and ASTM G94 (metals). Test piping and instrumentation should be suitable for pure oxygen environments. Sensors should provide safe and accurate data in pure oxygen environments.

6.3.5 Thermal compatibility. Candles generate oxygen flow at elevated temperatures. The oxygen flow will rise in temperature in the test facility piping and sensors during each burn. Temperature limitations of sensors are given by the manufacturers. Modifications to the test system piping or sensor location may be necessary to not exceed sensor temperature ranges. The activity performing the testing specified in section 4 should ensure flow temperature remains within recommended operating limits of the sensors.

6.3.6 Candle formulation additives. Research on the beneficial effect of various additives or catalysts are often published in scientific literature. These benefits include reducing hazardous materials and improving candle performance. Asbestos was once used as an additive to strengthen the candle as it burned. Asbestos is specifically forbidden in T9070-AL-DPC-020/077-2. Any requests for additives should be forwarded to the procuring activity.

6.3.7 Sensor selection. The following provides the typical values resulting from the various tests of section 4. The sensors selected to measure these tests during inspections should be capable of capturing the values provided, at a minimum.

6.3.7.1 Pressure sensor. Oxygen stream pressures before the flow meter are usually less than 5 psig (1.5 bar) but may vary depending on the type of flow meter. This is not the pressure in the furnace, which may become higher as the internal filters become coated with salt particles.

6.3.7.2 Flow meter. Oxygen flow rate during candle burn may rise significantly at the start and occasionally during the rest of the burn. The average flow may be as high as 3 standard cubic feet per minute (90 standard liters per minute). The specification requires the maximum instantaneous flow be less than five times the average or 15 standard cubic feet per minute (450 standard liters per minute).

6.3.7.3 Water vapor. During candle tests, the dew point of the oxygen stream may begin at the laboratory conditions and drop significantly during the burn. Dew point/freeze point temperatures of -15 °F (-25 °C) have been documented during historical candle burn tests. The average dew point may be computed in place of the average weight of water as a weighted average (see 4.6.1.5.1). Water vapor result may be presented as mass of water per volume of gas or as dew points.

6.3.7.4 Chlorine sensor. Typical chlorine concentrations during a candle test are below 1 ppm but may exceed 30 ppm at some point during the burn.

6.3.7.5 Carbon monoxide sensor. Carbon monoxide concentrations during a candle test may reach 300 ppm at some point during the burn.

6.3.7.6 Verification of sensors. Manufacturers' specification sheets for sensors should be provided to the Technical Authority for verification of performance characteristics prior to first article inspection.

6.3.8 Igniter construction. A 9- or 10-gauge foundry nail meets the dimensional requirements herein and is acceptable for use as an ignition device (see 3.6).

6.4 Standard test procedures. The test procedures described herein are the standard, and results obtained thereby should supersede all others. These procedures have been chosen because of their simplicity and historical success in ensuring chlorate candles used in the submarine fleet conform to the requirements of this specification.

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6.5 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.6 Supersession data. This specification supersedes CS-4460-0001, dated 6 May 2008, in its entirety, and all activities related to the subject matter should now reference this document for compliance with the latest requirements. National Stock Number 4460-00-875-4568, Chlorate Candles, was previously purchased for use on U.S. Navy Submarines in accordance with CS-4460-0001. All future procurements should be made in accordance with this specification.

6.7 Subject term (key word) listing.

Carbon monoxide
Chlorine
Flammable solid
Flow meter
Hazardous material
Igniter
Ignition zone
Key
Leak test
Oxidizer
Oxygen generator
Pressure transducer
Submarine
Water vapor

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4460-2026-003)

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